

Technological advancements in SMT machines

The SMT equipment market is witnessing major technological innovations. Growing sophistication and the shrinking size of electronic gadgets is fuelling these advancements. Recent demand within the industry has been for high-precision complex boards and assemblies. At the same time, there is a constant pressure to improve the quality of the goods produced, aiming for zero defects. To be able to manage this, manufacturers need to have the right SMT equipment, which will enable fast, flexible and perfect production of PCBs.

Nowadays, a wide range of machines are available. These offer throughput as high as 25,000-45,000 components per hour, with mount precision as good as 30 microns. High-speed changeover facilities, intelligent tape feeders and conveyors, 5-megapixel inline inspection systems, high-speed and high-resolution automatic optical inspection (AOI) systems, soldering robots, automatic high-speed printers with 12-second cycle time, 10-micron repeatable accuracies at fine pitch and six-channel profiler packs for reflows are some examples of the latest technology available today.

In order to cater to the demands of manufacturers, SMT equipment are undergoing non-stop technical advancements, and some of the significant trends that have been observed in the field are increased parallelisation and flexibility, high levels of software integration, environment-friendly machines for RoHS-based production, machines that can handle smaller components at a faster speed, and small yet intelligent machines.

India is to offer specialised engineering services as part of the PCB assembly introduction in the manufacturing floor. With the CEMs identifying many shortcomings in the OEM design in the post-manufacturing start, these are being introduced as engineering services. The CEMs look into the OEM design and highlight the possible additional tooling expenses due to violation of the design rule checklist that is drawn-up from the CEM's capability.

Failure analysis

The probability of device failures has increased substantially with the miniaturisation of SMDs. Also, failure analysis of failed devices has become a challenge than ever before. Specialised equipment and skillsets required for the purpose are yet to catch up in India.

Most of the failure analysis is done by respective component vendors outside India. However, fab-less semiconductor companies in India outsource failure analysis to third-party labs outside India. With more semiconductor companies coming up in India, a few such labs have opened their facilities within India. This has drastically reduced the turnaround time and enhanced correspondence within the engineering teams. This is a potential growth area supporting the electronics industry.

Standards

The growing demand for SMT in the electronics manufacturing segment in India has brought in the commitment from standardisation organisations like IPC, Electrostatic Discharge Association (ESDA) and JEDEC. IPC with its office in India has contributed significantly to awareness in the SMT domain through several international IPC training programmes. The most widely used assembly standard that covers IPC-A-610 E has been translated in Hindi.

As SMD packaging shrinks and stringent requirements need to be devised, ESDA and JEDEC are coming up with joint standards, which are being reviewed by committees from ESDA India. These activities help contribute to the international platform from Indian perspective, and incorporation of feedback encourages further progress and brainstorming in the developing areas.

Training

SMT training is available in India at different levels. Training in equipment installation, handling and operation, and facility set-up is provided by the respective equipment vendor. However, operator-level training for hands-on repair or rework and inspection can be had within India.

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